100-Class Image Classification via ResNet152

Visual Recognition HW1 — Technical Report

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Github repo: https://github.com/youzhe0305/NYCU-VRDL

1. Introduction

The task of this assignment is image classification: given an input image, the model must predict its corresponding category label. Specifically, this is a 100-class image classification problem. The dataset consists of 20,724 training images distributed across 100 classes, a validation set of 300 images, and 2,344 unlabeled test images for final prediction.

My core approach to this task is to apply transfer learning using a ResNet pretrained on ImageNet, and to progressively improve the model through systematic ablation experiments. This ultimately achieves 93.33% accuracy on the validation set.

2. Method

2.1 Model Architecture

The final model consists of three main components: a ResNet152 backbone, a GeM Pooling aggregation layer, and a lightweight linear classification head. The total parameter count is approximately 58.3M, within the 100M limit.

Backbone

The backbone is ResNet152 (He et al., 2016), initialized with ImageNet-1K V2 pretrained weights. ResNet152 was chosen because it achieves the best ImageNet performance among the ResNet variants reported in the original paper (He et al., 2016), while still fitting within the parameter constraint and being fast enough for iterative experimentation. Since the goal is transfer learning, all backbone layers are fully fine-tuned rather than frozen, so that important features specific to this dataset are not missed due to over-reliance on the pretrained representations.

GeM Pooling

We replace ResNet's standard Adaptive Average Pooling with Generalized Mean Pooling (GeM, Radenovic et al., 2019). The GeM pooling operation is defined as:

$$f = ((1/|X|) * sum(x_i ^ p)) ^ (1/p)$$

The exponent p is initialized to 3.0 and treated as a learnable parameter. When p = 1, GeM reduces to standard Average Pooling; when p → ∞, it approaches Max Pooling. This allows the model to learn how much emphasis to place on local discriminative features. In image classification, the key cues are typically localized — standard Average Pooling can dilute these important local features by mixing them with uninformative background regions, while GeM retains more discriminative information without discarding as much as Max Pooling would.

Classification Head

The classification head is kept minimal: Dropout(0.3) → Linear(2048, 100). Ablation experiments described later show that a deeper MLP head (e.g., 2048→512→100) performs worse, likely because extra layers introduce unnecessary complexity.

Module	Configuration
ResNet152 Backbone	ImageNet-1K V2 pretrained, all layers trainable
GeM Pooling	p=3.0 (learnable), replaces AvgPool
Dropout	p=0.3
Linear	2048 → 100
Total Parameters	58,348,709 (~58.3M)

Table 1: Final model architecture summary.

2.2 Training Strategy

Differential Learning Rates

The backbone and classification head are optimized with separate learning rates and weight decay values, balancing the preservation of pretrained knowledge against fast adaptation to the new task:

- Classification head: lr = 5e-4, weight_decay = 1e-4
- Backbone: lr = 5e-5, weight_decay = 1e-3

The higher backbone weight decay prevents catastrophic forgetting of generic ImageNet features. The lower backbone learning rate is especially important during the unstable early phase of training, shielding the pretrained weights from large disruptive updates.

Loss Function and Label Smoothing

We use cross-entropy loss with label smoothing (smoothing = 0.15). Without label smoothing, the model is trained to fit a one-hot label distribution, which drives it to maximize the gap between logits of different classes — increasing the risk of overfitting. Label smoothing distributes a small probability mass to non-target classes, discouraging over-confidence and improving generalization.

CutMix Regularization

CutMix (Yun et al., 2019) is applied with 50% probability per batch (alpha = 1). A rectangular region is cut from one image and pasted onto another, with labels mixed proportionally by area:

$$\tilde{x} = M \odot x_i + (1-M) \odot x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda) y_j$$

By requiring the model to simultaneously predict the labels of two images proportional to their visible area, CutMix encourages the model to learn calibrated, proportional logit responses rather than binary yes/no decisions. This prevents the model from over-committing to a class label based on only a small number of visible features, reducing misclassification from partial cues.

Learning Rate Scheduler

We use CosineAnnealingLR with T_max = 60 and eta_min = 1e-6. The learning rate decays smoothly from its initial value to near zero, enabling fast progress in early epochs and fine-grained convergence in later epochs.

2.3 Data Augmentation

Separate augmentation pipelines are applied for training and inference:

Augmentation	Training	Validation / Test
Resize / Crop	RandomResizedCrop(448, scale=(0.5, 1.0))	Resize(515) → CenterCrop(448)
Horizontal Flip	RandomHorizontalFlip(p=0.5)	—
Auto-Augment	TrivialAugmentWide()	—
Erasing	RandomErasing(p=0.25)	—
Normalize	ImageNet Mean / Std	ImageNet Mean / Std

Table 2: Data augmentation pipelines for training and inference.

Beyond standard flipping, erasing, and normalization, we increase the input resolution from ResNet's default 224×224 to 448×448, allowing the model to capture finer-grained details — a key factor contributing approximately +1.0% accuracy gain. TrivialAugmentWide (Müller & Hutter, 2021) uniformly samples a single transformation from a wide pool of operations at each step.

Because it applies only one transform per image, it avoids the risk of compounding multiple transformations that would over-distort the image and shift the distribution too far from the original, while still exposing the model to diverse visual variations to learn invariant features.

2.4 Hyperparameter Summary

Hyperparameter	Value
Optimizer	AdamW
Head Learning Rate	5e-4
Backbone Learning Rate	5e-5
Head Weight Decay	1e-4
Backbone Weight Decay	1e-3
LR Scheduler	CosineAnnealingLR (T_max=60, eta_min=1e-6)
Label Smoothing	0.15
Dropout	0.3
CutMix Alpha	1.0, applied with 50% probability
Batch Size	32
Input Resolution	448×448
Max Epochs	60
Mixed Precision (AMP)	Enabled (GradScaler)

Table 3: Final model hyperparameters.

3. Results

3.1 Best Model Performance

The final model achieves 93.33% accuracy on the 300-image validation set (280 correct predictions).

Metric	Value
Best Validation Accuracy	93.33% (280 / 300)
Best Epoch	23
Epochs Completed	60
Test Set Predictions	2,344 images

Table 4: Best model performance summary.

3.2 Cumulative Improvement Over Baseline

Starting from an 86.33% baseline (ResNet152 with basic augmentation), each design decision was evaluated independently through ablation experiments. The table below summarizes the cumulative accuracy at each improvement stage.

Improvement	Val Acc	Delta

Baseline (ResNet152 + basic aug.)	86.33%	—
+ TrivialAugmentWide	87.33%	+1.00%
+ Resolution 320×320	88.00%	+0.67%
+ CutMix (alpha=1.0)	90.33%	+2.33%
+ Dropout 0.5 → 0.3	91.00%	+0.67%
+ Resolution 448×448	92.00%	+1.00%
+ Batch size=32, LR tuning	93.00%	+1.00%
+ GeM Pooling	93.33%	+0.33%

Table 5: Cumulative accuracy improvement across design decisions.

4. Additional Experiments

This section summarizes the approaches explored through over 32 ablation experiments that were ultimately not adopted, along with analysis of why they did not improve performance.

4.1 Alternative Backbone: ResNet50

ResNet50 (~25M parameters) trains faster per epoch but its reduced model capacity and the lower diversity of features in its pretrained weights cap validation accuracy at approximately 89%, around 2% below ResNet152.

4.2 MixUp Augmentation

MixUp (Zhang et al., 2018) performs pixel-space linear interpolation between pairs of images. It showed no measurable improvement over the baseline. The blended images lack spatially coherent regions, providing limited signal for fine-grained label learning compared to CutMix, which preserves intact local patches with clear semantic content.

4.3 Multi-Layer MLP Classification Head (2048→512→100)

Adding an intermediate fully-connected layer (2048→512→100) performed worse than the single linear layer, with a drop of approximately 1.7%. The extra layer likely introduces unnecessary complexity without providing commensurate representational benefit.

4.4 Exponential Moving Average (EMA)

EMA maintains a shadow model as a running exponential average of the weights. Two implementation attempts both failed: one due to Batch Normalization running statistics not being updated for the EMA model, causing validation accuracy to collapse to ~2%, and another that

stabilized at only 71.7%. More fundamentally, EMA averaging smooths over the accuracy peak at epoch 23, pulling performance downward as training diverges after the peak.

4.5 Stochastic Weight Averaging (SWA)

SWA (Izmailov et al., 2018) averages checkpoints from the later training phase to find flatter loss minima. Since validation accuracy peaks at epoch 23 and declines afterward, averaging in lower-quality later checkpoints degrades the final model by approximately 0.3% compared to using the single best checkpoint.

4.6 RandAugment

RandAugment (Cubuk et al., 2020) applies N=2 randomly selected augmentations at a fixed magnitude M=9. It consistently underperformed TrivialAugmentWide by approximately 0.3%. TrivialAugmentWide applies only a single randomly chosen operation with a randomly sampled magnitude from a wider range, providing greater diversity while avoiding over-distortion from stacking multiple strong transforms.

4.7 Aggressive Random Resized Crop (scale=(0.08, 1.0))

Expanding the crop scale range to (0.08, 1.0) allows crops covering as little as 8% of the original image. Such extreme crops destroy most semantic content, making it difficult for the model to learn discriminative class features. This led to a ~3% drop in validation accuracy.

4.8 Summary

Method	Val Acc Change	Decision
Final Best Model (ResNet152 + GeM + CutMix)	93.33%	Adopted ✓
ResNet50 Backbone	−2%	Discarded
MixUp Augmentation	+0%	Discarded (limited signal for fine-grained learning)
Multi-Layer MLP Head (2048→512→100)	−1.7%	Discarded (unnecessary complexity)
EMA	−2.7%	Discarded (averages over peak)
SWA	−0.3%	Discarded (mixes in worse checkpoints)
RandAugment (N=2, M=9)	−0.3%	Discarded (worse than TrivialAugmentWide)
Aggressive Crop scale=(0.08, 1.0)	−3%	Discarded (semantic destruction)

Table 6: Summary of all additional experiments.

5. References

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